Inference for Models of Fish and Wildlife Population Dynamics

WILD(FISH) 8390
Lecture: Tues, Thurs 1:15–2:35 PM
Room 4-516, Warnell

	Instructor	Teaching Assistant
	Dr. Richard Chandler	Jen Brown
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Office hours:	Thur 2:45–3:45	Fri 11:30–1:30

Course Description

Recently developed statistical tools make it possible to fit models of population dynamics to field data while accounting for observation error. This course will explain how to use these tools for statistical inference and prediction about animal population parameters including abundance, occupancy, survival, recruitment, and dispersal.

Course Objectives and Learning Outcomes

By the end of the course, students will know how to (1) design studies of wildlife population dynamics, (2) fit statistical models to data collected in the field, (3) interpret results, and (4) make predictions useful for informing conservation decisions.

Textbooks

- (AHM1) Kéry, M. and J.A. Royle. 2016. Applied Hierarchical Modeling in Ecology: Analysis of Distribution, Abundance and Species Richness in R and BUGS : Volume 1: Prelude and Static Models. Academic Press. eBook.
- (AHM2) Kéry, M. and J.A. Royle. 2020. Applied Hierarchical Modeling in Ecology: Analysis of Distribution, Abundance and Species Richness in R and BUGS : Volume 2: Dynamic and Advanced Models. Academic Press. eBook.
- (SCR) Royle, J.A., R.B. Chandler, R. Sollmann, and B. Gardner. 2013. Spatial Capture-Recapture. Academic Press. Download chapters of the eBook from the UGA library eBook.

Grading

	Quantity	Grade percentage
Weekly assignments*	15	45%
Final presentation	1	20%
Final paper*	1	30%
Class participation		5%

*Late assignments will be penalized 3 points/day, up to a maximum of 50 points off.

The plus/minus grading system will be used, according to UGA policy, and assigned following this plus/minus grading scale: A = >93-100, A- = >90-93, B+ = >87-90, B = >83-87, B- = >80-83, C+ = >77-80, C = >73-77, C-= >70-73, D = 60-70, F = <60.

Attendance
Attendance is optional, but it will be difficult to succeed if you don't participate in lectures.

Academic Honesty
As a University of Georgia student, you have agreed to abide by the UGA academic honesty policy. UGA Student Honor code: "I will be academically honest in all of my academic work and will not tolerate academic dishonesty of others". A Culture of Honesty, the University's policy and procedures for handling cases of suspected dishonesty, can be found at here. You are responsible for informing yourself about the university's standards before performing any academic work. Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation. Please ask if you have questions related to course assignments and the academic honesty policy. Any form of possible academic dishonesty will be reported to the UGA Office of the Vice President for Instruction.

Generative AI
Artificial intelligence (AI) tools, like chatGPT and other generative AI chatbots, can be useful for improving text and computer code. However, they can also hinder learning if used to generate answers and bypass critical thinking. Students must disclose the use of generative AI for completing assignments, including the final paper. If a student decides to use generative AI, it can only be used to improve text and code written by the student. AI tools cannot be used to create answers by inputting assignment questions into prompts. Students must report the prompts provided to AI tools, along with the original text (or code) and the final text. Prompts and output can be disclosed by providing a link to the AI chat.

Academic Coaching
Assistance with time management, test and performance anxiety, notetaking, motivation, text comprehension, test preparation, and other barriers to success at UGA. Link for the Office of Academic Enhancement.

Accommodations for Disabilities
If you require a disability-required accommodation, it is essential that you register with the Disability Resource Center (Clark Howell Hall; 706-542-8719) and notify me of your eligibility for reasonable accommodations. We can then plan how best to coordinate your accommodations. Please note that accommodations cannot be provided retroactively.

UGA Well-Being Resources

UGA Well-being Resources promote student success by cultivating a culture that supports a more active, healthy, and engaged student community. Anyone needing assistance is encouraged to contact Student Care & Outreach (SCO) in the Division of Student Affairs at 706-542-8479 or visit `sco.uga.edu`. Student Care & Outreach helps students navigate difficult circumstances by connecting them with the most appropriate resources or services. They also administer the Embark@UGA program which supports students experiencing, or who have experienced, homelessness, foster care, or housing insecurity.

UGA provides both clinical and non-clinical options to support student well-being and mental health, any time, any place. Whether on campus, or studying from home or abroad, UGA Well-being Resources are here to help.

Student Care and Outreach: `sco.uga.edu`

University Health Center: `healthcenter.uga.edu`

Counseling & Psychiatric Services: `caps.uga.edu` or CAPS 24/7 crisis support 706-542-2273

Health Promotion/ Fontaine Center: `healthpromotion.uga.edu`

Accessibility and Testing: `accessibility.uga.edu`

Additional information, including free digital well-being resources, can be accessed through the UGA app or by visiting `https://well-being.uga.edu`.

Tentative Course Outline

Date	Lecture	Reading
	PART 1 – BACKGROUND	
Aug 18	Introduction and linear models	AHM1 Foreword, Preface, Chapter 1
Aug 20	Generalized linear models	AHM1 Chapter 3
	PART 2 – STATIC MODELS	
Aug 25	Occupancy models	AHM1 Ch 4 and Ch 10 (pp 551–564)
Aug 27	Occupancy models	
Sept 1	Occupancy models	AHM1 Chapter 10 (pp 564–600)
Sept 3	Occupancy models	AHM1 Chapter 10 (pp 591–600)
Sept 8	Binomial N-mixture models	AHM1 Chapter 6 (pp 219–228)
Sept 10	Prior predictive checks	AHM1 Chapter 6 (pp 229–245)
Sept 15	Model selection and fit assessment	AHM1 Chapter 6 (pp 245–254)
Sept 17	Causal inference	AHM1 Chapter 6 (pp 254–312)
Sept 22	Multinomial N-mixture models	AHM1 Chapter 7 (pp 313–334)
Sept 24	Multinomial N-mixture models	AHM1 Chapter 7 (pp 334–391)
Sept 29	Hierarchical distance sampling	AHM1 Chapter 8 (pp 426–460)
Oct 1	Conventional distance sampling	AHM1 Chapter 8 (pp 393–426)
Oct 6	Non-spatial capture-recapture	SCR Chapters 1–4
Oct 8	Spatial capture-recapture	SCR Chapter 5
Oct 13	Spatial capture-recapture	SCR Chapter 7
Oct 15	Spatial capture-recapture	SCR Chapters 9
	PART 3 – DYNAMIC MODELS	
Oct 20	Dynamic occupancy models	AHM2 Ch4 (Due: first draft)
Oct 22	Dynamic N-mixture models	AHM2 Chapter 2
Oct 27	Survival models	AHM2 Chapter 3
Oct 29	Cormack-Jolly-Seber models	SCR Chapter 16
Nov 3	Jolly-Seber models	SCR Chapter 16 (Due: peer review)
Nov 5	Jolly-Seber models	SCR Chapter 16
Nov 10	Dynamic spatial capture-recapture models	SCR Chapter 16
Nov 12	Dynamic spatial capture-recapture models	SCR Chapter 16
Nov 17	Student presentations	
Nov 19	Student presentations	
Nov 24	Thanksgiving Break	
Nov 26	Thanksgiving Break	
Dec 1	No class (Friday schedule)	Due: Final paper

The course syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.